

e) $\csc^2 x + \frac{3}{2} \csc x - 1 = 0$

$2 \csc^2 x + 3 \csc x - 2 = 0$

$(2 \csc x - 1)(\csc x + 2) = 0$

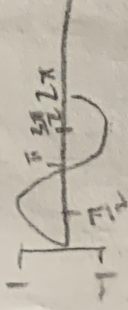
case ①

$\csc x = \frac{1}{2}$

$\sin x = 2$

2 is out of

the range of
 $\sin x$ (range is $y \in [-1, 1]$)



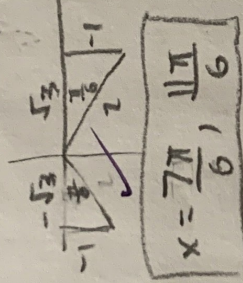
∴ No solutions

case ②

$\csc x = -2$

$\sin x = -\frac{1}{2}$

RAA = $\frac{\pi}{6}$



g) $\sin^2 x - \sqrt{2} \cos x = \cos^2 x + \sqrt{2} \cos x + 2$

$(1 - \cos^2 x) - \sqrt{2} \cos x = \cos^2 x + \sqrt{2} \cos x + 2$

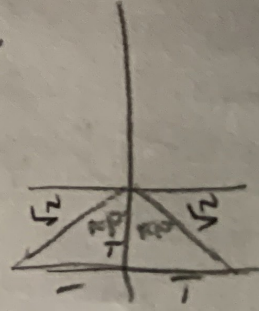
$2 \cos^2 x + 2\sqrt{2} \cos x + 1 = 0$

$\cos x = \frac{-2\sqrt{2} \pm \sqrt{(2\sqrt{2})^2 - 4(2)(1)}}{2(2)}$

$\cos x = \frac{-2\sqrt{2} \pm \sqrt{0}}{4}$

$\cos x = -\frac{1}{\sqrt{2}}$

RAA = $\frac{\pi}{4}$



f) $-2 \cos 2x = 2 \sin x$

$-(\cos 2x) = \sin x$

$-(1 - 2 \sin^2 x) = \sin x$

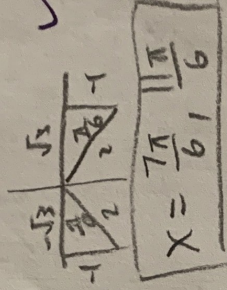
$2 \sin^2 x - \sin x - 1 = 0$

$(2 \sin x + 1)(\sin x - 1) = 0$

case ①

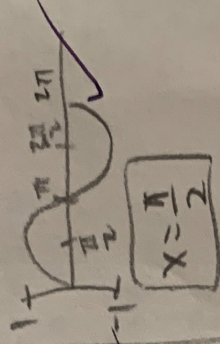
$\sin x = -\frac{1}{2}$

RAA = $\frac{\pi}{6}$



case ②

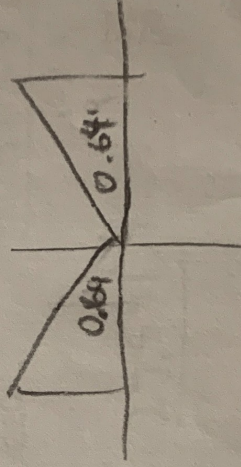
$\sin x = 1$



h) $\sin 2x - 0.6 = 0$

$\sin 2x = 0.6$

RAA = 0.64



$2x = 0.64, 2.50, 6.93, 8.78$

$x = 0.32, 1.25, 3.46, 4.39$

1. Solve each equation for x , where $0 \leq x \leq 2\pi$. Leave in exact radian measure where possible otherwise round to two decimal places.

a) $\cot x \csc^2 x = 2 \cot x$

$\cot x \csc^2 x - 2 \cot x = 0$

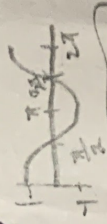
$\cot x (\csc^2 x - 2) = 0$

Case ①

$\cot x = 0$

$\frac{\cos x}{\sin x} = 0$

$\cos x = 0$



$x = \frac{\pi}{2}, \frac{3\pi}{2}$

Case ②

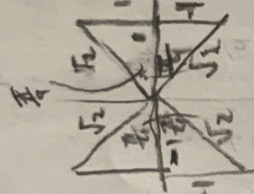
$\csc^2 x - 2 = 0$

$\csc^2 x = 2$

$\sin^2 x = \frac{1}{2}$

$\sin x = \pm \frac{1}{\sqrt{2}}$

$RAA = \frac{\pi}{4}$
 $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$



c) $8 \sin^2 x - 8 \sin x + 1 = 0$

$\sin x = \frac{8 \pm \sqrt{8^2 - 4(8)(1)}}{2(8)}$

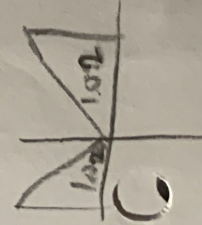
$= \frac{1 \pm \sqrt{2}}{4}$

$= \frac{2 \pm \sqrt{2}}{4}$

Case ①

$\sin x = \frac{2 + \sqrt{2}}{4}$

$RAA = 1.023$

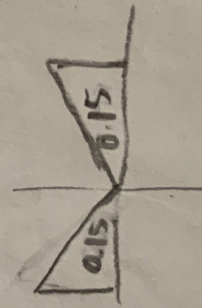


$x = 1.023, 2.12$

Case ②

$\sin x = \frac{2 - \sqrt{2}}{4}$

$RAA = 0.15$



$x = 0.15, 2.99$

b) $3 \tan^3 x - \tan x = 0$

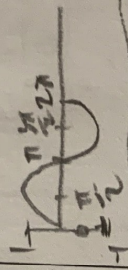
$\tan x (3 \tan^2 x - 1) = 0$

Case ①

$\tan x = 0$

$\frac{\sin x}{\cos x} = 0$

$\sin x = 0$



$x = 0, \pi, 2\pi$

Case ②

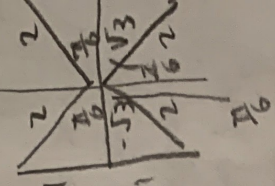
$3 \tan^2 x - 1 = 0$

$\tan^2 x = \frac{1}{3}$

$\tan x = \pm \frac{1}{\sqrt{3}}$

$RAA = \frac{\pi}{6}$

$x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$



d) $5 \cos 2x - \cos x + 3 = 0$

$5(2 \cos^2 x - 1) - \cos x + 3 = 0$

$10 \cos^2 x - 5 - \cos x + 3 = 0$

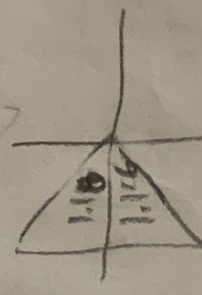
$10 \cos^2 x - \cos x - 2 = 0$

$(5 \cos x + 2)(2 \cos x - 1) = 0$

Case ①

$\cos x = -\frac{2}{5}$

$RAA = 1.106$

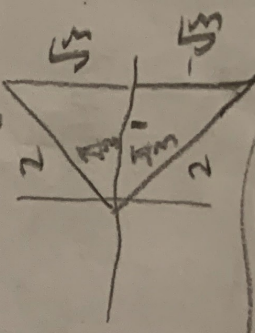


$x = 1.98, 4.30$

Case ②

$\cos x = \frac{1}{2}$

$RAA = \frac{\pi}{3}$



$x = \frac{\pi}{3}, \frac{5\pi}{3}$